Chapter 22: Magnetism

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PHY201

Lecture 22

Part I

Magnets and Magnetic Fields

Outline of Part I: Magnets and Magnetic Fields

Outline of Part I: Magnets and Magnetic Fields

What is Magnetism?

Magnetism is a force that acts at a distance, attracting or repelling certain materials, produced by *moving electric charges*.

Everyday Examples

- Compass needles align with Earth's field
- Refrigerator magnets
- Electric motors and generators
- MRI scanners
- Hard drives and speakers

Magnetism and electricity are **two aspects of a single force**: electromagnetism.

A stationary charge creates only an electric field.

A *moving* charge creates both an electric field *and* a magnetic field.

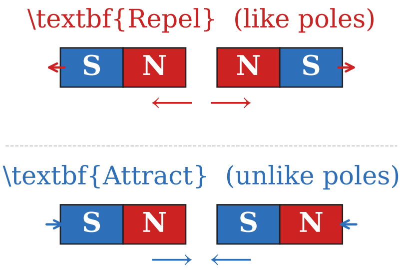
Magnetic Poles

Every Magnet Has Two Poles

- **North (N)** pole
- **South (S)** pole
- Like poles repel
- Unlike poles attract

No Magnetic Monopoles!

Cutting a magnet in half does *not* produce isolated N and S poles.
Each piece becomes a smaller complete magnet.

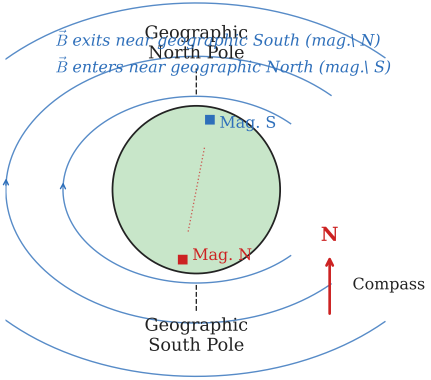


Earth as a Giant Magnet

Earth's Magnetic Dipole

- Earth behaves like a giant bar magnet
- Geographic North Pole ≈ Earth's magnetic **south** pole
- Compass N end points toward geographic north because *opposite poles attract*
- Magnetic axis is tilted $\sim 11.5°$ from rotation axis

The field is generated by convection currents of liquid iron in Earth's outer core.



Check Your Understanding

A bar magnet is broken in half. What happens?

If the north end of a compass points toward geographic north, what type of magnetic pole is near geographic north?

Come to lecture to see the answer.

Outline of Part I: Magnets and Magnetic Fields

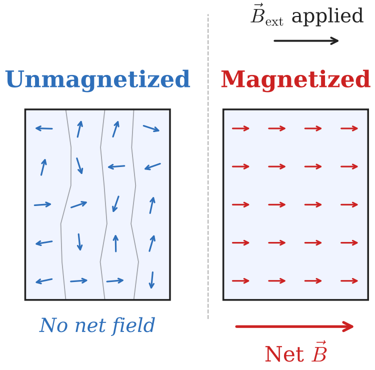
Ferromagnetism and Magnetic Domains

Ferromagnetic Materials

Strongly attracted to magnets:
iron, cobalt, nickel (and alloys)

Magnetic Domains

- Microscopic regions — each acts as a tiny magnet
- **Unmagnetized**: domains point randomly $\Rightarrow$ no net field
- **Magnetized**: external field aligns domains $\Rightarrow$ strong net field
- Above the **Curie temperature**, thermal agitation randomizes domains; ferromagnetism disappears



Magnetism from Electric Current

Oersted's Discovery (1820)

Hans Christian Oersted discovered that a current-carrying wire deflects a nearby compass needle — proving that **electric current produces a magnetic field**.

All Magnetism = Moving Charge

- Electron **orbital motion** creates tiny magnetic dipoles
- Electron **spin** also contributes
- In ferromagnets, these align to form domains

Electromagnets

Current through a coil of wire creates a temporary magnet that can be switched on/off — used in motors, MRI, cranes

- **Soft** ferromagnets demagnetize easily (e.g. temporary electromagnet cores)
- **Hard** ferromagnets retain magnetization (e.g. permanent magnets)

Check Your Understanding

Why does iron gradually lose its magnetization after an external magnet is removed?

Come to lecture to see the answer.

Outline of Part I: Magnets and Magnetic Fields

The Magnetic Field $\vec{B}$

Magnetic Field — SI Unit: tesla (T)

- The magnetic field $\vec{B}$ is a vector field describing the magnetic influence at each point in space
- Direction: the direction a compass needle's **north end** points
- $1\,\mathrm{T} = 1\,\frac{\mathrm{N}}{\mathrm{A\,m}}$
- Also common: $1\,\mathrm{G} = 10^{-4}\,\mathrm{T}$ (gauss)

Typical Field Strengths

- Earth's field: $\sim 5 \times 10^{-5}\,\mathrm{T}$
- Refrigerator magnet: $\sim 0.01\,\mathrm{T}$
- MRI scanner: 1–3 T
- Strong research magnet: $\sim 45\,\mathrm{T}$

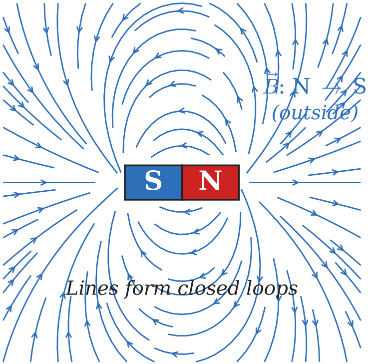
$\vec{B}$ is analogous to $\vec{E}$

but has fundamentally different sources and behavior.

Magnetic Field Lines

Properties of Field Lines

- Tangent to the line gives $\vec{B}$ direction
- Closer spacing $\Rightarrow$ stronger field
- Lines **never cross**
- Lines always form **closed loops** (no monopoles)
- Outside a magnet: N$\rightarrow$S
- Inside a magnet: S$\rightarrow$N



Unlike electric field lines (which begin/end on charges), magnetic field lines have no beginning or end.

Check Your Understanding

What does closely spaced magnetic field lines indicate?

Outside a bar magnet, in which direction do magnetic field lines run?

Come to lecture to see the answer.

Part II

Magnetic Forces on Moving Charges

Outline of Part II: Magnetic Forces on Moving Charges

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Magnetic Force on a Moving Charge

Circular Motion in Magnetic Fields

Applications: Velocity Selector and Mass Spectrometer

The Hall Effect

Magnetic Force: $F = qvB\sin\theta$

Magnetic Force on a Moving Charge — SI Unit: newton (N)

A charge q moving with speed v in a magnetic field B at angle θ to $\vec{B}$ experiences:

$$\boxed{F = qvB\sin\theta}$$

Variable Definitions

- q: charge (C)
- v: speed $\left(\frac{\mathrm{m}}{\mathrm{s}}\right)$
- B: magnetic field magnitude (T)
- θ: angle between $\vec{v}$ and $\vec{B}$

Special Cases

- $\theta = 90°$: **maximum force** $F = qvB$
- $\theta = 0°$ or $180°$: **zero force**
- Charge parallel to $\vec{B}$: no magnetic force

Magnetic force **never** does work on a charge!

Direction: Right-Hand Rule 1 (RHR-1)

RHR-1 for Positive Charges

1. Point fingers along $\vec{v}$
2. Curl fingers toward $\vec{B}$
3. Thumb points in direction of $\vec{F}$

For Negative Charges

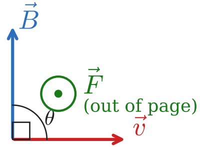
Reverse the direction of $\vec{F}$ (or use the left hand).

$\vec{F}$ is always **perpendicular** to both $\vec{v}$ and $\vec{B}$.

This is why the magnetic force does no work: $W = \vec{F} \cdot \vec{d} = 0$.

Right-Hand Rule 1 (RHR-1)

For $-q$: reverse $\vec{F}$ (use left hand)



\textbf{3.} Thumb $\Rightarrow$ $\vec{F}$ direction

\textbf{2.} Curl toward $\vec{B}$

\textbf{1.} Point fingers along $\vec{v}$

Check Your Understanding

A particle moves to the right in a magnetic field directed out of the page. The magnetic force on the particle is directed *downward*. What is the sign of the charge?

Come to lecture to see the answer.

Example: Force on a Proton — (cp2e_ch22_ex1)

Force on a Proton in Earth's Field

A proton moves eastward at $v = 7.5 \times 10^6\ \frac{\text{m}}{\text{s}}$ through Earth's magnetic field $B = 5.0 \times 10^{-5}\,\text{T}$, directed north. Find the magnetic force on the proton.

$$q = 1.60 \times 10^{-19}\,\text{C}$$

$$\theta = 90^\circ \quad (\vec{v} \perp \vec{B})$$

$$F = qvB\sin\theta$$

$$= (1.60 \times 10^{-19})(7.5 \times 10^6)(5.0 \times 10^{-5})\sin 90^\circ$$

$$\boxed{F = 6.0 \times 10^{-17}\,\text{N}}$$

Direction by RHR-1:
$\vec{v}$ east, $\vec{B}$ north $\Rightarrow$ $\vec{F}$ **upward** (away from Earth's surface).

- This tiny force is why Van Allen belts trap charged particles.
- Compare to gravity: $mg \approx 1.6 \times 10^{-27}\,\text{N}$ — negligible.

Outline of Part II: Magnetic Forces on Moving Charges

Circular Motion: Magnetic Force as Centripetal Force

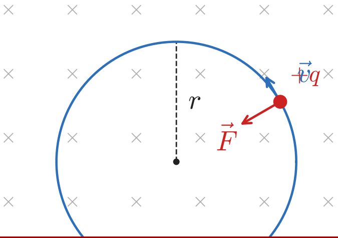
When $\vec{v} \perp \vec{B}$: Circular Orbit

The magnetic force is always perpendicular to velocity $\Rightarrow$ it acts as a **centripetal force**:

$$qvB = \frac{mv^2}{r} \qquad \Rightarrow \qquad \boxed{r = \frac{mv}{qB}}$$

Physical Interpretation

- Larger m or $v \Rightarrow$ larger orbit
- Larger q or $B \Rightarrow$ smaller orbit
- Speed is *constant* (no work done)
- Direction changes continuously



Example: Electron Radius of Curvature — (cp2e_ch22_ex2)

Electron in a Magnetic Field

An electron moves perpendicular to $\vec{B} = 0.500\,\mathrm{T}$ at $v = 6.00 \times 10^{7}\,\frac{\mathrm{m}}{\mathrm{s}}$. Find its radius of curvature.

$$m = 9.11 \times 10^{-31}\,\mathrm{kg}$$

$$q = 1.60 \times 10^{-19}\,\mathrm{C}$$

$$r = \frac{mv}{qB}$$

$$= \frac{(9.11 \times 10^{-31})(6.00 \times 10^{7})}{(1.60 \times 10^{-19})(0.500)}$$

$$\boxed{r = 6.83 \times 10^{-4}\,\mathrm{m}}$$

$r = 0.683\,\mathrm{mm}$

The very small radius shows how magnetic fields tightly steer electron beams in CRTs, accelerators, and electron microscopes.

- Does the speed change? No — magnetic force is perpendicular to $\vec{v}$.
- What doubles r? Doubling v or halving B.

Check Your Understanding

At equal speed in the same magnetic field, which particle follows the larger circular arc: a proton or an electron? Explain.

Come to lecture to see the answer.

Outline of Part II: Magnetic Forces on Moving Charges

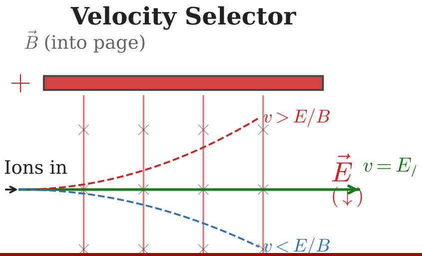
Velocity Selector

Crossed Electric and Magnetic Fields

When $\vec{E}$ and $\vec{B}$ are perpendicular, a particle travels straight only when the electric and magnetic forces exactly cancel:

$$qE = qvB \qquad \Rightarrow \qquad \boxed{v = \frac{E}{B}}$$

- Only particles with speed $v = E/B$ pass straight through
- Faster particles deflect in one direction
- Slower particles deflect in the other
- Result: a **monochromatic beam** of particles



Mass Spectrometer

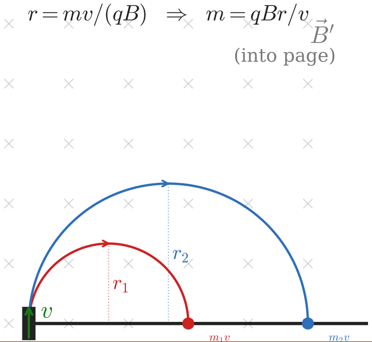
Separating Isotopes by Mass

After velocity selection, particles enter a region with $\vec{B}$ only and follow circular arcs of radius $r = mv/(qB)$.

Different masses $\Rightarrow$ different radii $\Rightarrow$ different detector positions.

Applications

- Isotope identification and separation
- Environmental and forensic analysis
- Drug testing
- Carbon-14 dating
- Detecting trace elements in geology



Outline of Part II: Magnetic Forces on Moving Charges

The Hall Effect

Charge Separation in a Magnetic Field

When current flows through a conductor in a perpendicular magnetic field, charge carriers are deflected sideways. This creates a **transverse voltage** called the **Hall voltage** (or Hall EMF):

$$\boxed{\varepsilon = B\ell v}$$

where ℓ is the width of the conductor and v is the drift speed.

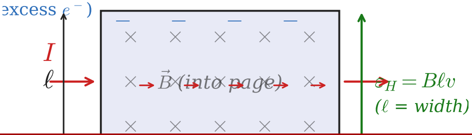
Derivation Logic

At equilibrium, electric restoring force = magnetic deflecting force:

$$qE = qvB \;\Rightarrow\; E = vB$$

$$\varepsilon = E\ell = B\ell v$$

Hall Effect



Hall Effect Applications

Why the Hall Effect Matters

- Determines the **sign of charge carriers** in a material
- In most metals: electrons (negative carriers)
- In some semiconductors: holes (positive carriers)

Hall Probe Applications

- Measuring magnetic field strength (Hall probe)
- Non-invasive blood flow measurement
- Position sensors in motors and joysticks

Reversing current direction *reverses* the Hall voltage — this is how the sign of the charge carrier is identified.

$\varepsilon = B\ell v$ is very small in metals but measurable with sensitive electronics.

Example: Blood Flow Measurement — (cp2e_ch22_ex3)

Hall Effect Blood Flow Sensor

A Hall effect sensor is used to measure blood flow. Given $B = 0.100\,\mathrm{T}$, vessel width $\ell = 4.00\,\mathrm{mm}$, and blood speed $v = 20.0\,\frac{\mathrm{cm}}{\mathrm{s}}$, find the Hall EMF.

$$\varepsilon = B\ell v$$

$$= (0.100\,\mathrm{T})(4.00 \times 10^{-3}\,\mathrm{m})(0.200\,\tfrac{\mathrm{m}}{\mathrm{s}})$$

$$\boxed{\varepsilon = 8.00 \times 10^{-5}\,\mathrm{V} = 80.0\,\mu\mathrm{V}}$$

$80\,\mu\mathrm{V}$ is tiny but detectable with modern electronics.

This technique is **non-invasive** and gives a direct measure of blood speed — used clinically for cardiac output measurements.

Check Your Understanding

If the magnetic field in a Hall effect sensor is doubled, what happens to the Hall voltage?

Why does reversing the current direction reverse the Hall voltage?

Come to lecture to see the answer.

Part III

Magnetic Forces on Currents

Outline of Part III: Magnetic Forces on Currents

Outline of Part III: Magnetic Forces on Currents

Force on a Current-Carrying Conductor

Torque on a Current Loop: Motors

Magnetic Force on a Wire

Force on a Current-Carrying Wire — SI Unit: newton (N)

A wire of length ℓ carrying current I in a magnetic field B at angle θ experiences:

$$\boxed{F = I\ell B\sin\theta}$$

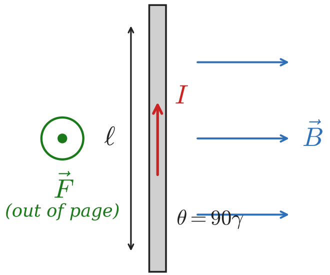
Derivation Idea

Each charge carrier experiences $F = qvB\sin\theta$. Adding over all carriers in length ℓ: current $I = nqv_dA$, summing gives $F = I\ell B\sin\theta$.

Direction

Use RHR-1 with:
thumb $\to$ current direction, **fingers** $\to$ $\vec{B}$ $\Rightarrow$ force on wire.

Force on Current-Carrying Wire



Example: Force on a Power Line — (cp2e_ch22_ex4)

Force on a Current-Carrying Wire

A wire carries $I = 20.0\,\mathrm{A}$ over a length $\ell = 0.0500\,\mathrm{m}$ in a field $B = 1.50\,\mathrm{T}$ perpendicular to the wire. Find the force on the wire.

$$\theta = 90^\circ$$

$$F = I\ell B \sin\theta$$

$$= (20.0\,\mathrm{A})(0.0500\,\mathrm{m})(1.50\,\mathrm{T}) \sin 90^\circ$$

$$\boxed{F = 1.50\,\mathrm{N}}$$

$1.50\,\mathrm{N}$ is a substantial force on a short wire — this is the principle behind electric motors and loudspeakers.

- If the wire were parallel to $\vec{B}$, $F = 0$.
- Maximum force when current $\perp$ field.

Check Your Understanding

A current-carrying wire is placed in a magnetic field parallel to the current. What is the magnetic force on the wire?

If the wire is then rotated to be perpendicular to the field, how does the force change?

Come to lecture to see the answer.

Outline of Part III: Magnetic Forces on Currents

Torque on a Current Loop

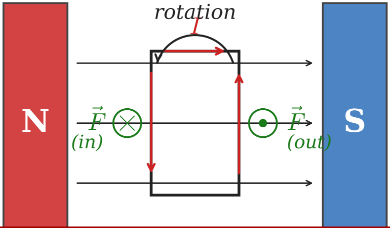
Torque on a Current Loop — SI Unit: $\mathrm{N \cdot m}$

A loop of N turns, area A, carrying current I in field B at angle θ between the area vector and $\vec{B}$:

$$\boxed{\tau = NIAB\sin\theta}$$

How the Torque Arises

- Opposite sides of the loop carry current in opposite directions
- Forces on these sides are equal and opposite
- This creates a **couple** (pure torque) tending to align the loop with $\vec{B}$
- Maximum torque when loop plane is



Electric Motors and Galvanometers

The Electric Motor Principle

- Torque on the loop causes rotation
- Without intervention, the loop aligns with $\vec{B}$ and torque stops
- A **commutator** reverses current direction every half-turn
- This maintains torque and sustains rotation

Galvanometer

- A coil in a radial magnetic field
- Deflection $\propto$ current
- Basis of analog ammeters and voltmeters

$\tau_{\text{max}} = NIAB$

Occurs when the loop plane is parallel to $\vec{B}$ (area vector $\perp$ $\vec{B}$).

Increasing N, I, A, or B all increase the torque and motor power.

Example: Motor Torque — (cp2e_ch22_ex5)

Torque on a Motor Coil

A motor coil has $N = 100$ turns, side length $0.100\,\mathrm{m}$, current $I = 15.0\,\mathrm{A}$, and field $B = 2.00\,\mathrm{T}$. Find the maximum torque.

$$A = (0.100)^2 = 0.0100\,\mathrm{m}^2$$

$$\tau = NIAB\sin\theta \quad (\theta = 90^\circ \Rightarrow \max)$$

$$= (100)(15.0)(0.0100)(2.00)$$

$$\boxed{\tau = 30.0\,\mathrm{N\cdot m}}$$

$30.0\,\mathrm{N\cdot m}$ is a substantial torque for a compact device.

Industrial motors achieve large torques by using many turns, strong magnets, and high currents.

- What doubles the torque? Double N, I, A, or B.
- When is $\tau = 0$? When the loop is

Part IV

Magnetic Fields from Currents

Outline of Part IV: Magnetic Fields from Currents

Outline of Part IV: Magnetic Fields from Currents

Right-Hand Rule 2: Fields from Currents

RHR-2 for Current-Generated Magnetic Fields

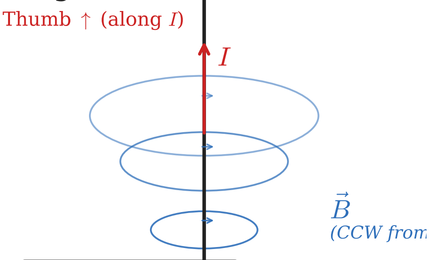
1. Point your **thumb** in the direction of conventional current
2. Your **curled fingers** show the direction of the magnetic field lines circling the wire

Field lines form concentric circles around a long straight wire.

Field direction *reverses* if current direction reverses.

The field is strongest near the wire and decreases with distance.

Right-Hand Rule 2 (RHR-2)



\textbf{RHR-2:} Thumb $\parallel I$
Fingers curl → direction of $\vec{B}$

Magnetic Field of a Long Straight Wire

Long Straight Wire — SI Unit: tesla (T)

The magnetic field at distance r from a long straight wire carrying current I:

$$\boxed{B = \frac{\mu_0 I}{2\pi r}}$$

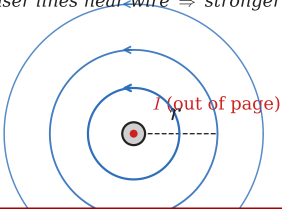
where $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T \cdot m/A}$ is the **permeability of free space**.

Key Features

- Field decreases as $1/r$ (not $1/r^2$!)
- Field lines are concentric circles
- Doubling current doubles B
- Doubling distance halves B

Long Straight Wire

Denser lines near wire $\Rightarrow$ stronger $\vec{B}$



Example: Field Near a Wire — (cp2e_ch22_ex6)

Magnetic Field from a Long Straight Wire

A long straight wire carries $I = 25\,\text{A}$. Find the magnetic field at $r = 5.00\,\text{cm}$ from the wire.

$$B = \frac{\mu_0 I}{2\pi r}$$

$$= \frac{(4\pi \times 10^{-7})(25\,\text{A})}{2\pi(5.00 \times 10^{-2}\,\text{m})}$$

$$\boxed{B = 1.00 \times 10^{-4}\,\text{T}}$$

$B = 1.00 \times 10^{-4}\,\text{T} = 1.00\,\text{G}$

This is about twice Earth's magnetic field — a substantial field close to a high-current wire.

- Direction: use RHR-2 with thumb along current.
- Note: 2π cancels, simplifying the calculation.

Solenoids: Uniform Magnetic Fields

Solenoid Field — SI Unit: tesla (T)

Inside a solenoid with $n = N/\ell$ turns per unit length carrying current I:

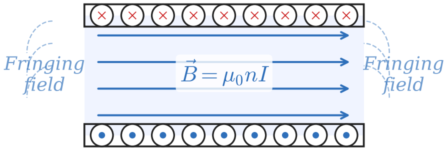
$$\boxed{B = \mu_0 n I}$$

The field inside is **nearly uniform and parallel to the axis.**

- Many loops superimpose to produce a strong, uniform field
- Field outside is nearly zero
- Analogous to a uniform electric field inside a capacitor
- Used in: MRI, particle accelerators, electromagnets

Solenoid Field

× (into page) \\ · (out of page): current loops



Example: MRI Solenoid — (cp2e_ch22_ex7)

Field Inside a Solenoid

An MRI solenoid has $n = 1000\,\frac{1}{\mathrm{m}}$ turns per meter and carries $I = 2.00\,\mathrm{kA}$. Find the field inside.

$$B = \mu_0 n I$$

$$= (4\pi \times 10^{-7})(1000\,\tfrac{1}{\mathrm{m}})(2000\,\mathrm{A})$$

$$\boxed{B = 2.51\,\mathrm{T}}$$

2.51 T is a typical clinical MRI field.

Such strong, uniform fields are required to align hydrogen nuclei for nuclear magnetic resonance imaging.

- Doubling n doubles B.
- Field is uniform throughout the bore regardless of position.

Check Your Understanding

A long straight wire carries $I = 10\,\mathrm{A}$. At what distance from the wire is $B = 4.0 \times 10^{-5}\,\mathrm{T}$ (approximately Earth's field)?

Come to lecture to see the answer.

Outline of Part IV: Magnetic Fields from Currents

Force Between Parallel Current-Carrying Wires

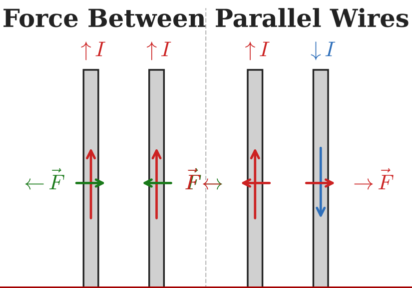
Force per Unit Length Between Parallel Wires

Two parallel wires separated by distance r, carrying currents I_1 and I_2:

$$\boxed{\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r}}$$

Direction Rules

- **Parallel** (same direction) currents: **attract**
- **Antiparallel** (opposite direction) currents: **repel**
- Note: this is the *opposite* of what you might expect from charges!



Example: Force Between Parallel Wires — (cp2e_ch22_ex8)

Attraction Between Parallel Wires

Two parallel wires are separated by $r = 0.0400\,\mathrm{m}$ and carry $I_1 = 5.00\,\mathrm{A}$ and $I_2 = 3.00\,\mathrm{A}$ in the same direction. Find the force per unit length between them.

$$\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

$$= \frac{(4\pi \times 10^{-7})(5.00)(3.00)}{2\pi(0.0400)}$$

$$\boxed{\frac{F}{\ell} = 7.50 \times 10^{-5}\,\frac{\mathrm{N}}{\mathrm{m}}}$$

Force is **attractive** (same-direction currents).

$75\,\frac{\mu\mathrm{N}}{\mathrm{m}}$ — small but measurable; important in power systems where large bus bars carry parallel currents.

Check Your Understanding

Two parallel wires carry currents in opposite directions. Do they attract or repel?

If both currents are doubled, how does the force per unit length change?

Come to lecture to see the answer.

Part V

Applications of Magnetism

Outline of Part V: Applications of Magnetism

Outline of Part V: Applications of Magnetism

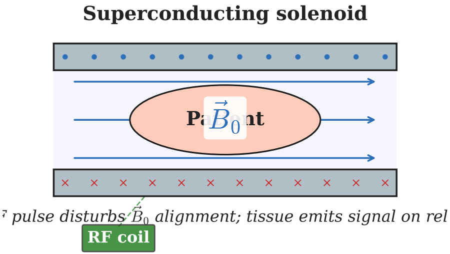
More Applications of Magnetism

Magnetic Resonance Imaging (MRI)

How MRI Works

- Superconducting solenoid produces 1–3 T uniform field
- Strong $\vec{B}$ aligns hydrogen nuclei (protons) in the body
- A radio-frequency (RF) pulse tips the protons out of alignment
- As protons relax back, they emit RF signals
- Different tissues relax at different rates $\Rightarrow$ image contrast

MRI Schematic



MRI: Advantages and Physics

Advantages over X-rays

- **Non-ionizing**: uses radio waves, not X-rays
- Excellent **soft-tissue contrast**
- Multiplanar imaging (axial, sagittal, coronal)
- No bone shadow artifacts

Limitations

- Expensive and slow scan times
- Contraindicated with metal implants
- Less effective for bone imaging

The solenoid field must be extremely uniform (< 1 ppm variation) across the imaging volume.

Superconducting coils cooled with liquid helium maintain current indefinitely with no power input.

Chapter 22 Summary

Key Equations

$$F = qvB\sin\theta$$

$$r = \frac{mv}{qB}$$

$$\varepsilon = B\ell v$$

$$F = I\ell B\sin\theta$$

$$\tau = NIAB\sin\theta$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \mu_0 nI$$

$$\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

Big Ideas

- Magnetism arises from **moving charges**
- Magnetic force $\perp \vec{v} \Rightarrow$ no work, no speed change
- Currents create fields; fields exert forces on currents
- Magnetic interactions depend strongly on **geometry** (angle θ)

Applications: motors, Hall sensors, mass spectrometers, MRI, Van Allen belts, particle accelerators.

Example: Velocity Selector Condition — (cp2e_ch22_ex9)

Velocity Selector

A velocity selector has $E = 1.50 \times 10^4\,\frac{\mathrm{V}}{\mathrm{m}}$ and $B = 0.0300\,\mathrm{T}$. What speed of particles passes straight through?

$$v = \frac{E}{B}$$

$$= \frac{1.50 \times 10^4\,\frac{\mathrm{V}}{\mathrm{m}}}{0.0300\,\mathrm{T}}$$

$$\boxed{v = 5.00 \times 10^5\,\frac{\mathrm{m}}{\mathrm{s}}}$$

Only particles with exactly $v = 5.00 \times 10^5\,\frac{\mathrm{m}}{\mathrm{s}}$ experience equal and opposite electric and magnetic forces and travel straight through.

All other speeds are deflected and blocked.